



AI-Powered Behavioural Threat Detection and SOC Risk Intelligence

Honeywell Hackathon 2026

Report

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Hackathon: Honeywell Hackathon 2026

DATE: 26/7/26

ABSTRACT

The rapid digital transformation of enterprises has significantly increased the complexity of cybersecurity threats. Modern organizations generate millions of authentication logs, network events, access requests, and endpoint activities every day. Security analysts are expected to identify malicious behaviour from this massive volume of data in real time. Traditional Intrusion Detection Systems (IDS) and Security Information and Event Management (SIEM) solutions primarily rely on predefined attack signatures or manually configured detection rules. Although these approaches are effective for previously known attacks, they often fail to detect insider threats, zero-day exploits, credential theft, and advanced persistent attacks where malicious behaviour closely resembles legitimate user activity.

SentinelAI is an Artificial Intelligence driven behavioural cybersecurity platform designed to overcome these limitations. Instead of identifying attacks based on fixed signatures, the system learns the normal behavioural patterns of users and devices using a Transformer-based deep learning model. Daily enterprise activities such as login behaviour, resource access, geographical locations, session duration, command sequences, and download patterns are analysed to determine whether a user's behaviour deviates significantly from their historical profile.

The proposed system combines unsupervised behavioural anomaly detection with deterministic cybersecurity rules to generate comprehensive threat intelligence. For every employee-day activity, SentinelAI produces an anomaly score, assigns a risk level, classifies the probable attack type, maps it to the MITRE ATT&CK framework, provides explainable reasoning behind the decision, and presents all findings through an interactive Security Operations Center dashboard. The platform also incorporates advanced capabilities such as cold-start handling, concept drift adaptation, multi-stage kill-chain correlation, and analyst-friendly investigation workflows.

Unlike conventional AI models that directly predict attack labels, SentinelAI intentionally separates behavioural detection from attack classification. The Behavioural Transformer focuses only on learning normal behaviour, while deterministic cybersecurity logic performs risk assessment and attack categorization. This architecture eliminates label leakage, improves transparency, and makes the system more suitable for real-world enterprise deployment.

The complete platform consists of synthetic enterprise data generation, behavioural feature engineering, Transformer-based anomaly detection, risk assessment, explainability, FastAPI backend services, and a React-based Security Operations Center dashboard, providing an end-to-end intelligent cybersecurity monitoring solution.

Keywords

Cybersecurity, Behavioural Analytics, Artificial Intelligence, Transformer Networks, Anomaly Detection, Security Operations Center, MITRE ATT&CK, Explainable AI, FastAPI, React.

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CHAPTER 1

INTRODUCTION

Cybersecurity has become one of the most critical challenges faced by modern enterprises. Organizations depend heavily on digital infrastructure for communication, cloud computing, financial transactions, employee collaboration, and customer services. As a result, enterprise environments continuously generate enormous volumes of security-related events including authentication logs, network requests, VPN connections, endpoint activities, database queries, and cloud access records.

The responsibility of analysing these events lies with the Security Operations Center (SOC), where analysts monitor thousands of alerts every day. Unfortunately, traditional monitoring systems depend primarily on predefined signatures or manually configured rules. While such approaches successfully identify previously known attacks, they are unable to detect sophisticated threats that imitate legitimate user behaviour.

Modern cyber attackers rarely launch noisy attacks. Instead, they frequently exploit stolen credentials, perform slow lateral movement, access sensitive resources gradually, and blend their activities into normal enterprise operations. Because these attacks appear legitimate from a traditional security perspective, they often remain undetected until significant damage has already occurred.

Artificial Intelligence has emerged as a promising solution for addressing these limitations. Instead of searching for known attack signatures, AI systems learn the normal behavioural patterns of users and devices over time. Any deviation from this learned behaviour can then be treated as a potential security anomaly for further investigation.

SentinelAI has been developed based on this behavioural security paradigm. The platform continuously analyses enterprise activities, learns behavioural patterns using a Transformer-based deep learning architecture, detects suspicious deviations, calculates intelligent risk scores, maps attacks to the MITRE ATT&CK framework, and provides explainable evidence through an interactive dashboard. The goal is not merely to generate alerts but to assist SOC analysts in making informed decisions with greater confidence and reduced investigation time.

1.1 Problem Statement

Enterprise cybersecurity systems face several practical challenges that reduce their effectiveness against modern attacks. Signature-based detection mechanisms are incapable of recognising unknown threats because they depend entirely on previously observed attack patterns. Similarly, insider threats and credential theft often appear indistinguishable from legitimate user behaviour, making them difficult to detect using conventional security tools.

Another significant challenge is alert fatigue. Modern SIEM platforms generate thousands of alerts every day, many of which are false positives. Security analysts spend valuable time investigating harmless events while genuine threats may remain unnoticed. Behavioural changes over time also create difficulties because static detection rules cannot adapt to evolving user habits. Additionally, new employees and newly registered devices possess limited historical data, making anomaly detection unreliable during the initial observation period. These challenges demonstrate the need for an intelligent behavioural monitoring system capable of learning normal user activity, identifying suspicious deviations, adapting to behavioural evolution, and providing explainable threat assessments to security analysts.

1.2 Objectives

The primary objective of SentinelAI is to develop an end-to-end behavioural cybersecurity platform capable of detecting sophisticated enterprise threats without relying exclusively on predefined attack signatures.

The specific objectives of the proposed system are:

- Learn normal behavioural patterns of enterprise users and devices using a Transformer-based deep learning model.
- Detect behavioural anomalies through unsupervised sequence modelling.
- Calculate dynamic risk scores based on behavioural evidence and security rules.
- Classify detected anomalies into meaningful cyberattack categories.
- Map identified attacks to the MITRE ATT&CK framework for standardized incident analysis.
- Generate explainable AI outputs to assist SOC analysts during investigations.
- Support cold-start handling for newly observed entities.
- Adapt to evolving behavioural patterns through concept drift detection.
- Correlate multiple suspicious events across different days into complete attack campaigns.
- Provide an interactive dashboard for enterprise security monitoring.

1.3 Scope of the Project

SentinelAI provides a complete behavioural cybersecurity pipeline beginning with enterprise event collection and ending with analyst-friendly threat investigation. The system accepts enterprise logs generated from authentication systems, VPN gateways, endpoint devices, application servers, and cloud resources. These events undergo feature engineering to construct behavioural profiles representing each employee's daily activities.

The Behavioural Transformer analyses these behavioural sequences and assigns anomaly scores indicating how much the observed behaviour deviates from historical patterns. Additional modules perform cold-start adjustment, concept drift adaptation, deterministic risk assessment, attack classification, MITRE ATT&CK mapping, and explainability generation. Finally, all results are displayed through an interactive React-based Security Operations Center dashboard served by a FastAPI backend.

Although the current implementation uses synthetic enterprise datasets for experimentation and evaluation, the overall architecture has been designed to integrate seamlessly with real-world enterprise SIEM solutions and cloud-based logging infrastructures.

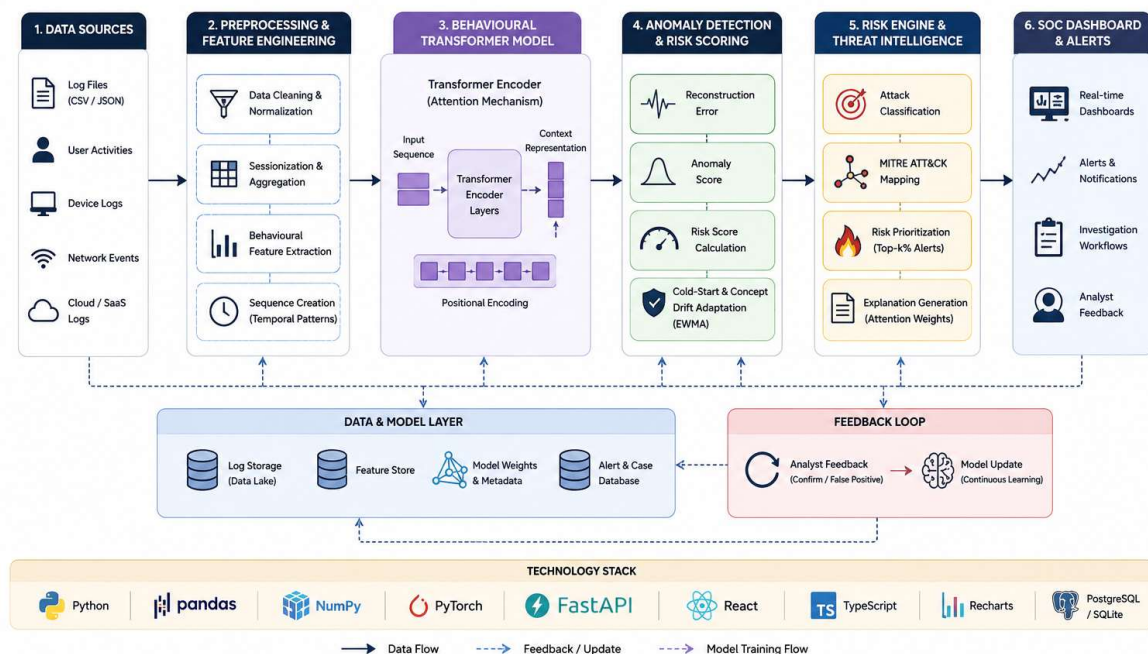
1.4 Applications

SentinelAI can be deployed across multiple cybersecurity domains where behavioural monitoring and intelligent threat detection are required.

Some potential applications include:

- Enterprise Security Operations Centers (SOC)
- Cloud Infrastructure Security Monitoring
- Insider Threat Detection
- Financial Institution Fraud Monitoring
- Government Cyber Defence
- Healthcare Information Security
- Critical Infrastructure Protection
- Industrial Control Systems (ICS)
- Corporate Network Security
- Identity and Access Management (IAM)

Figure 1.1: Overall System Workflow



CHAPTER 2

EXISTING SYSTEM

Modern enterprises primarily rely on Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Security Information and Event Management (SIEM) platforms, and Endpoint Detection and Response (EDR) solutions to monitor security events. These systems analyse network traffic, authentication logs, endpoint activities, and application logs using predefined signatures, heuristics, or manually configured detection rules. While effective against known attacks, they struggle to identify novel threats, insider attacks, and behavioural anomalies.

Traditional systems operate by comparing incoming events against a database of known attack patterns. If a match is found, an alert is generated. However, attackers continuously modify their techniques to evade detection, making signature-based approaches increasingly ineffective. Moreover, organisations generate millions of events daily, resulting in a large number of false positives that overwhelm Security Operations Center (SOC) analysts.

Machine learning has been introduced in recent years to improve detection accuracy. However, many existing AI-based solutions are supervised, requiring labelled attack datasets for training. Since real-world cyberattacks are rare and constantly evolving, obtaining comprehensive labelled datasets is difficult. This limits the ability of supervised models to generalise to previously unseen threats.

Additionally, most existing systems lack explainability, making it difficult for analysts to understand why an alert was generated. They also struggle with behavioural evolution over time and often fail to handle new users or devices with limited historical data.

2.1 Limitations of Existing Systems

The following limitations motivate the development of SentinelAI:

- Dependence on predefined attack signatures.
- Poor detection of zero-day attacks.
- Limited capability to identify insider threats.
- High false positive rates causing alert fatigue.
- Inability to adapt to changing user behaviour.
- Lack of explainability for AI-generated alerts.
- Difficulty handling new employees and devices (cold-start problem).
- Limited support for correlating multiple low-risk events into a complete attack campaign.

CHAPTER 3

PROPOSED SYSTEM

SentinelAI is an end-to-end behavioural cybersecurity platform that detects threats by learning how users and devices normally behave rather than relying on predefined attack signatures. The system combines unsupervised deep learning with deterministic cybersecurity rules to provide intelligent threat detection, risk assessment, and explainable incident analysis.

The proposed architecture separates anomaly detection from attack classification. A Behavioural Transformer model learns normal activity patterns and produces anomaly scores without using attack labels. These anomaly scores are then processed by a rule-based Risk Engine that evaluates behavioural indicators such as failed login attempts, unusual geographic locations, excessive downloads, after-hours access, VPN usage, and resource access entropy.

Based on these observations, SentinelAI assigns a risk score, determines the attack category, maps the incident to the MITRE ATT&CK framework, and generates human-readable explanations for SOC analysts. The platform also performs cold-start handling for new entities, concept drift adaptation for evolving user behaviour, and kill-chain correlation to connect multiple suspicious events into a single investigation case.

This architecture ensures transparency, scalability, and adaptability while reducing false positives and improving the efficiency of enterprise security operations.

3.1 Advantages of the Proposed System

The proposed system offers several advantages over conventional cybersecurity monitoring solutions:

- Learns behavioural patterns without requiring labelled attack data.
- Detects previously unseen attacks and insider threats.
- Reduces false positives through behavioural analysis.
- Provides explainable AI outputs for analyst confidence.
- Adapts to changing behaviour using concept drift detection.
- Handles newly observed users through cold-start adjustment.
- Correlates multi-stage attacks using kill-chain analysis.
- Supports batch and near real-time predictions.
- Integrates seamlessly with enterprise SOC workflows.

CHAPTER 4

SYSTEM MODULES

This chapter describes each functional module of SentinelAI. Every module contributes to the overall behavioural threat detection pipeline and works together to provide an intelligent Security Operations Center (SOC) solution. The screenshots included in this chapter demonstrate the implementation of each module and the interaction flow between different components of the system.

4.1 Landing Page Module

The Landing Page serves as the public-facing interface of SentinelAI. It introduces the platform, highlights its core capabilities, and provides visitors with an overview of the behavioural cybersecurity solution before accessing the Security Operations Center (SOC).

Developed using **Next.js**, the landing page follows a modern, responsive design and presents the key features of SentinelAI, including AI-powered behavioural anomaly detection, explainable threat analysis, MITRE ATT&CK mapping, and real-time SOC monitoring. It also provides navigation to the live demo, project documentation, and GitHub repository.

The landing page is designed to give recruiters, judges, and potential users a clear understanding of the product's purpose and architecture without requiring access to the internal dashboard.

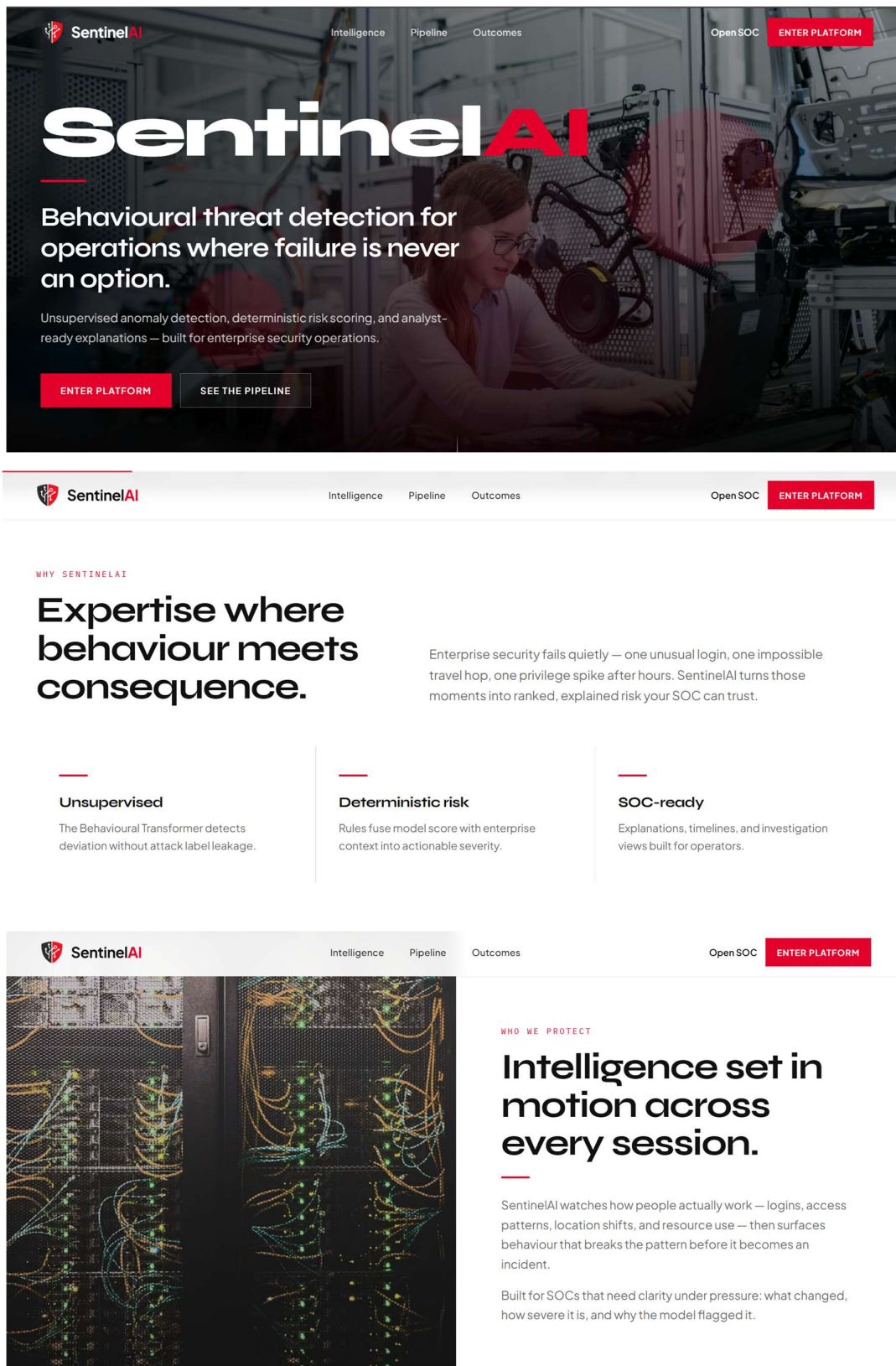
Key Features

- Responsive user interface
- Product overview and feature highlights
- Navigation to SOC Dashboard
- Technology stack showcase
- Project architecture overview
- Live Demo and GitHub links
- Contact and project information

Technologies Used

- Next.js
- React
- Tailwind CSS
- TypeScript
- Framer Motion

Figure 4.1 Landing Page




Intelligence
Pipeline
Outcomes
Open SOC
ENTER PLATFORM

HOW IT WORKS

From raw activity to explained risk — in one continuous pipeline.

Detection stays unsupervised. Risk and explanations stay deterministic. No attack labels leak into production scoring.

01

Feature engineering

Timeline events become numerical behavioural vectors — session rhythm, access entropy, location velocity, and more.

02

Anomaly detection

A Behavioural Transformer scores unsupervised deviation from each employee's normal operating baseline.

03

Risk fusion

Deterministic rules fuse anomaly signal with enterprise context into severity analysts can act on.

04

Explainability

Every finding ships with plain-language drivers — so investigations start with answers, not hunches.


Intelligence
Pipeline
Outcomes
Open SOC
ENTER PLATFORM

OUTCOMES

Tangible results for security teams under real operational load.

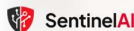
Built for environments where missed signals carry real consequence — refineries of identity data, hospitals of privilege, and plants of continuous access.



01

Anomaly clarity

See which sessions break the behavioural baseline — without drowning in raw event noise.


Intelligence
Pipeline
Outcomes
Open SOC
ENTER PLATFORM

02

Risk that ranks

Severity fused from model score and enterprise rules so the queue always starts with what matters.





03

Investigations that move

4.2 Upload Module

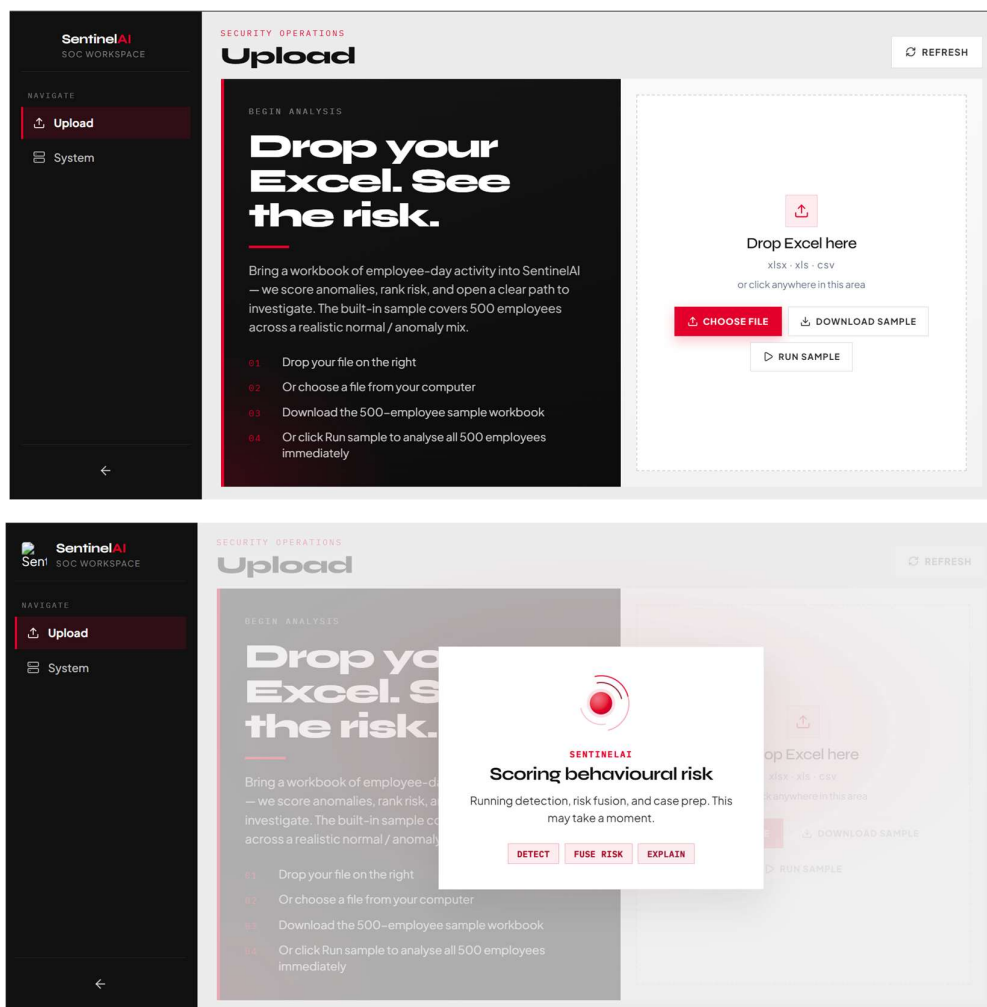
The Upload Module acts as the entry point of SentinelAI. It enables SOC analysts to upload enterprise log files in CSV or Excel format for behavioural analysis. The module also provides an option to analyse preloaded sample datasets, allowing quick demonstrations without requiring external data.

After a dataset is uploaded, it is validated and forwarded to the backend API, where feature engineering and behavioural analysis are initiated. The uploaded data is then processed by the Behavioural Transformer model and subsequent security modules.

Features

- Upload CSV and Excel files
- Sample dataset option
- Automatic validation
- Batch prediction support
- FastAPI integration

Figure 4.2 Upload Module



4.3 Dashboard Overview Module

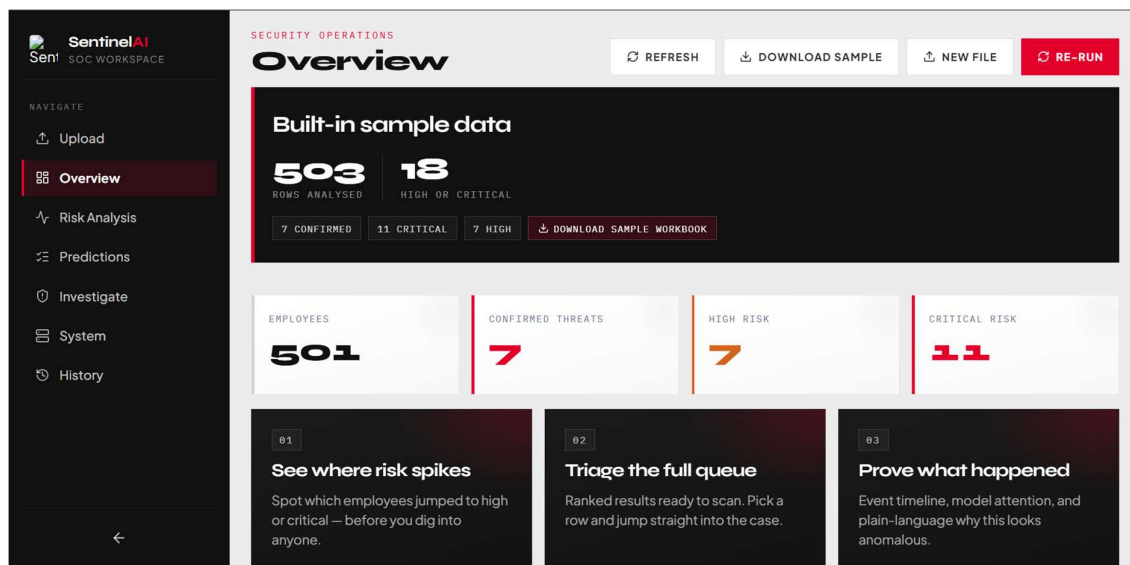
The Dashboard Overview provides a high-level summary of the uploaded dataset and the current security posture of the enterprise. It presents key statistics such as the total number of analysed users, detected anomalies, confirmed threats, and overall risk distribution.

The dashboard enables analysts to quickly assess organisational security and identify areas requiring immediate attention.

Displays

- Total Users Analysed
- Total Alerts
- Confirmed Threats
- Average Risk Score
- Security Status Summary

Figure 4.3 Dashboard Overview



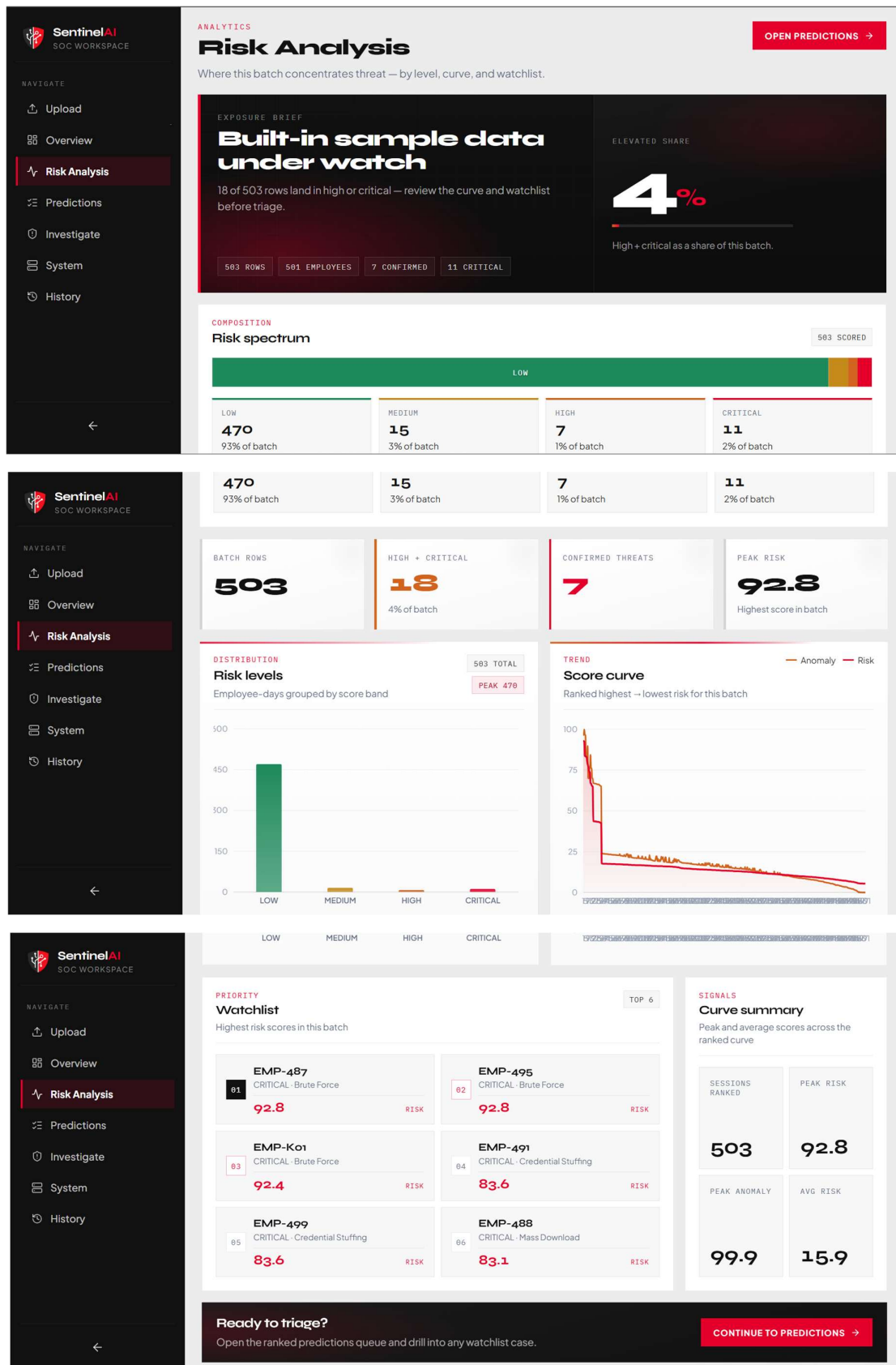
4.3 Risk Analysis Module

The Risk Analysis Module converts anomaly scores into meaningful risk levels. Each analysed user is assigned a numerical risk score ranging from 0 to 100, which is categorised into Low, Medium, High, or Critical.

Risk calculation considers multiple behavioural indicators such as authentication failures, unusual locations, abnormal download activity, VPN usage, and session behaviour.

Risk Levels: Low, Medium, High, Critical

Figure 4.3 Risk Analysis Dashboard



4.4 Predictions Module

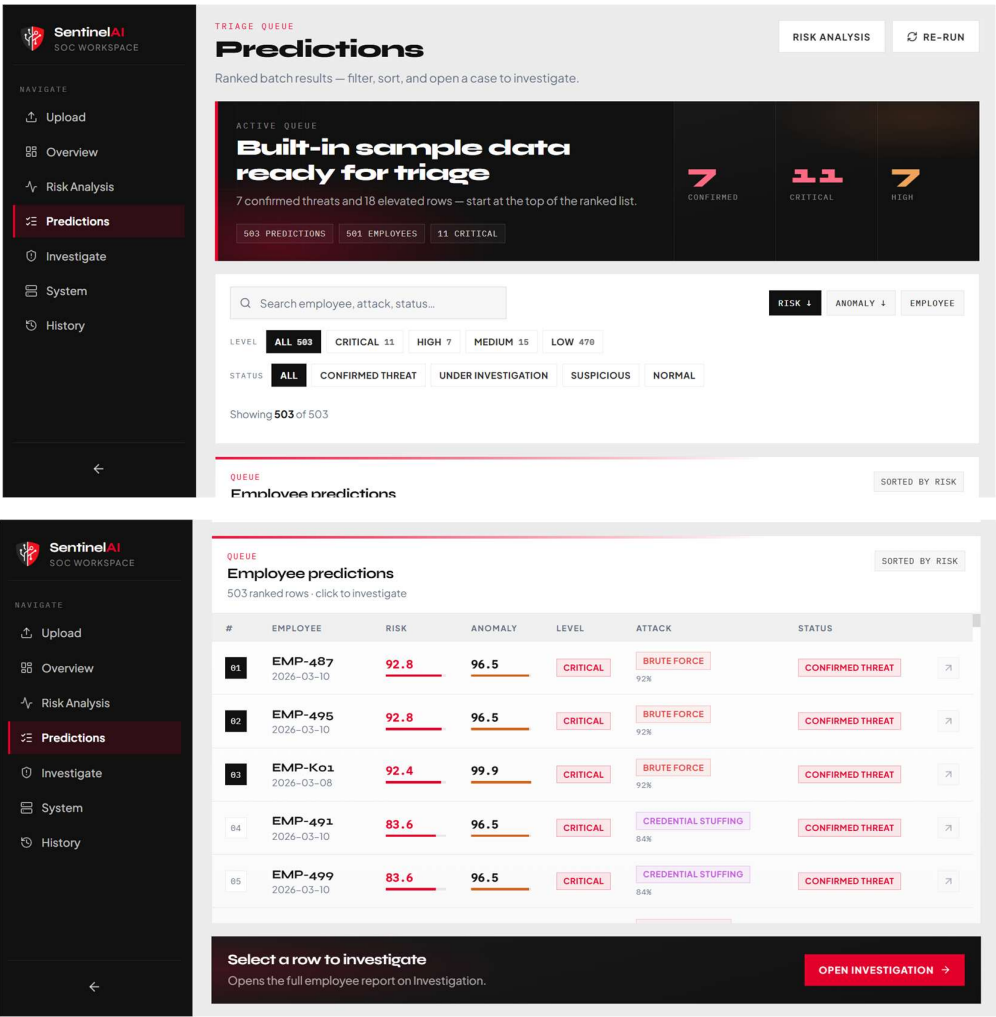
The Predictions Module displays the results generated by the Behavioural Transformer. Each row represents an analysed employee or device and includes anomaly score, attack type, confidence level, and current investigation status.

Analysts can filter, search, and prioritise alerts according to organisational requirements.

Information Displayed

- Employee ID
- Risk Score
- Behavioural Status
- Attack Type
- Confidence
- Recommendation

Figure 4.4 Predictions Page



4.5 Investigation Module

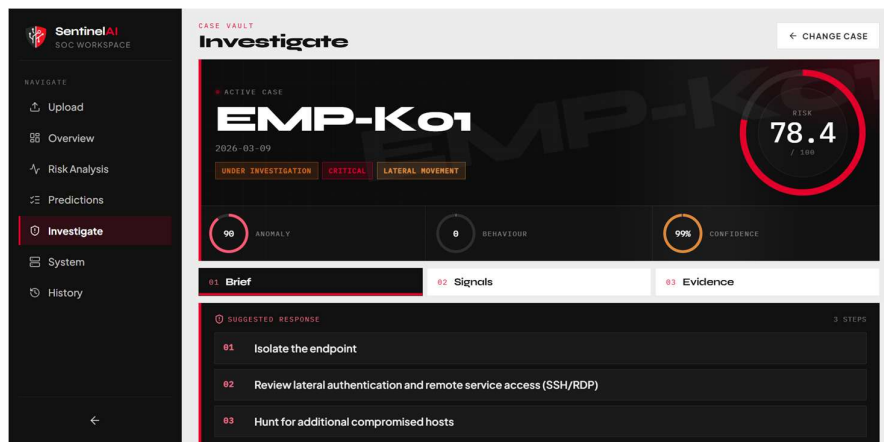
The Investigation Module provides detailed information for every suspicious case. It consolidates anomaly scores, behavioural explanations, MITRE ATT&CK mapping, attack classification, and analyst recommendations into a single investigation interface.

This module significantly reduces investigation time by presenting all relevant evidence together.

Includes

- Behaviour Summary
- Risk Assessment
- MITRE Mapping
- Triggered Indicators
- Recommendation
- Timeline

Figure 4.5 Investigation Module



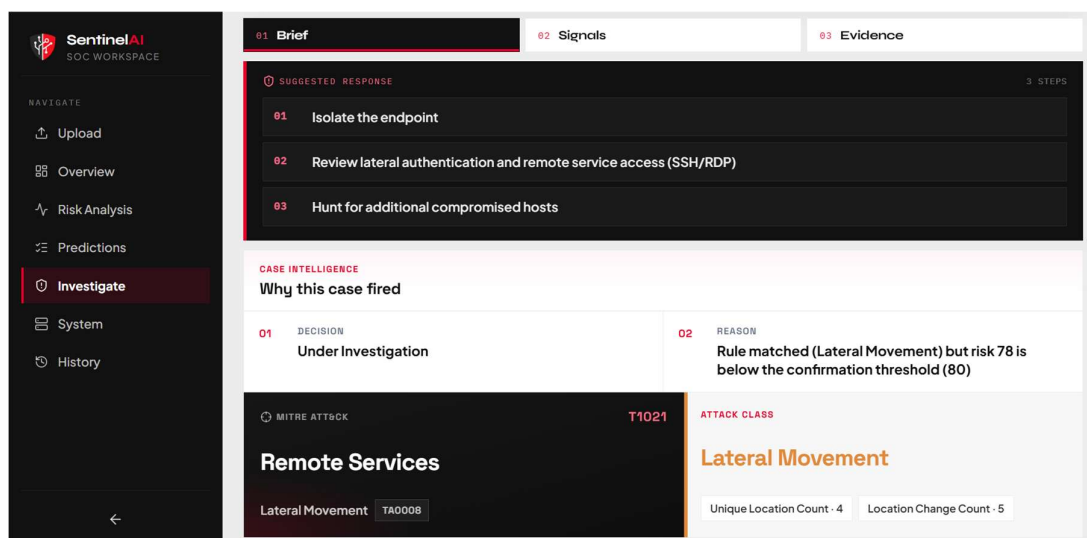
4.6 Explainability Module

The Explainability Module helps analysts understand why an alert was generated. Instead of presenting only numerical scores, the module highlights the behavioural factors responsible for the anomaly and explains how the final decision was reached.

Displays

- Behaviour Summary
- Contributing Factors
- Model Attention
- Risk Explanation
- Analyst Recommendation

Figure 4.6 Explainability Module

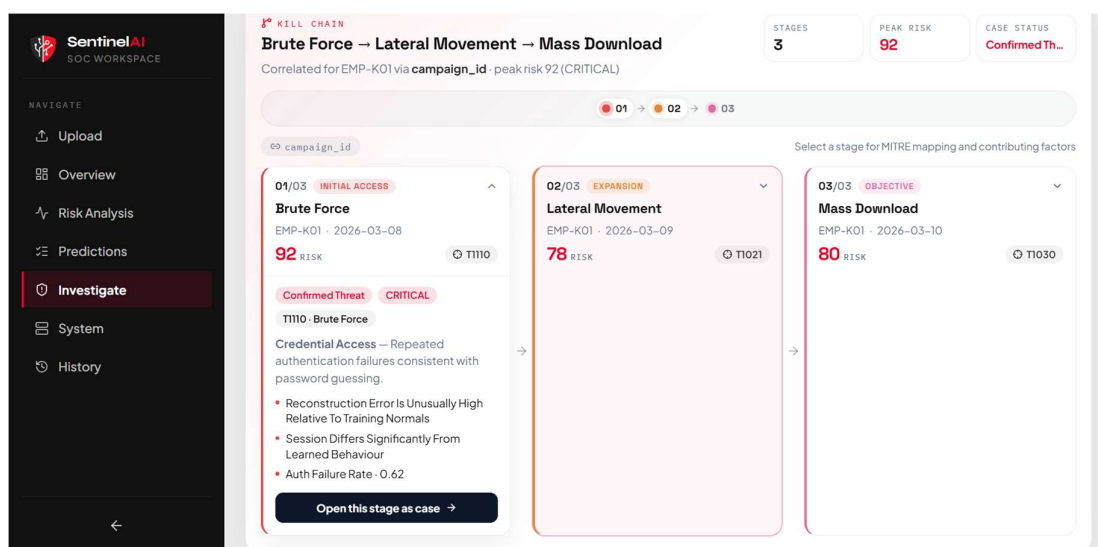


4.7 Kill-Chain Correlation Module

One of the novel features of SentinelAI is its ability to correlate multiple suspicious events into a single attack campaign. Rather than treating each alert independently, the module identifies relationships between events occurring across multiple days and constructs a complete attack timeline.

This enables analysts to understand the progression of complex attacks, such as brute force attempts followed by lateral movement and data exfiltration.

Figure 4.7 Kill-Chain Correlation



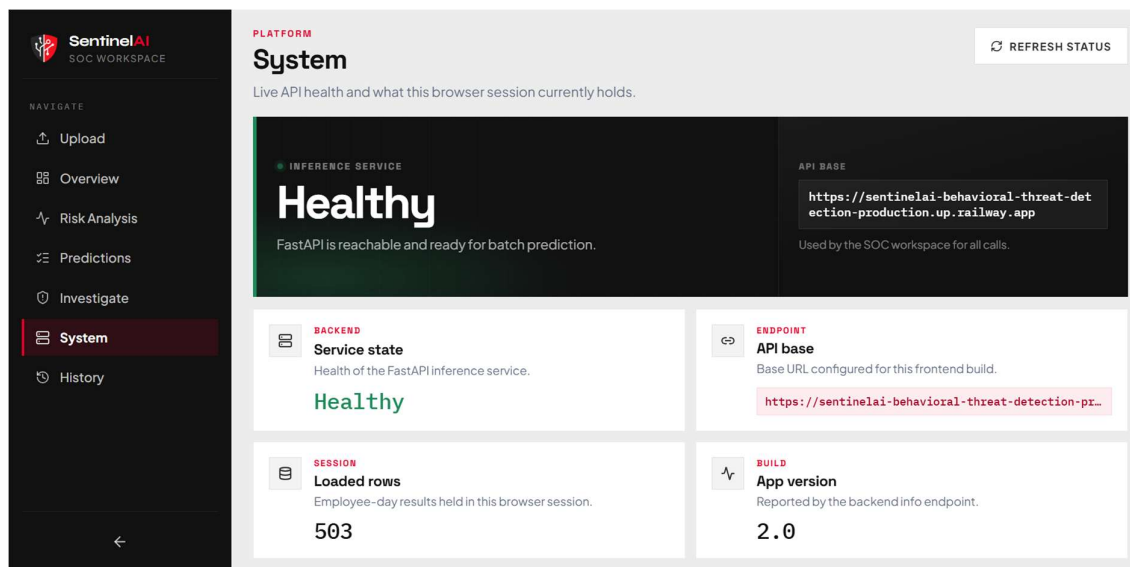
4.8 System Health Module

The System Health Module monitors the operational status of the SentinelAI platform. It displays backend connectivity, API response status, model availability, and system performance metrics to ensure reliable operation.

Displays

- Backend Status
- API Health
- Model Loaded Status
- Response Time
- Active Sessions

Figure 4.8 System Health



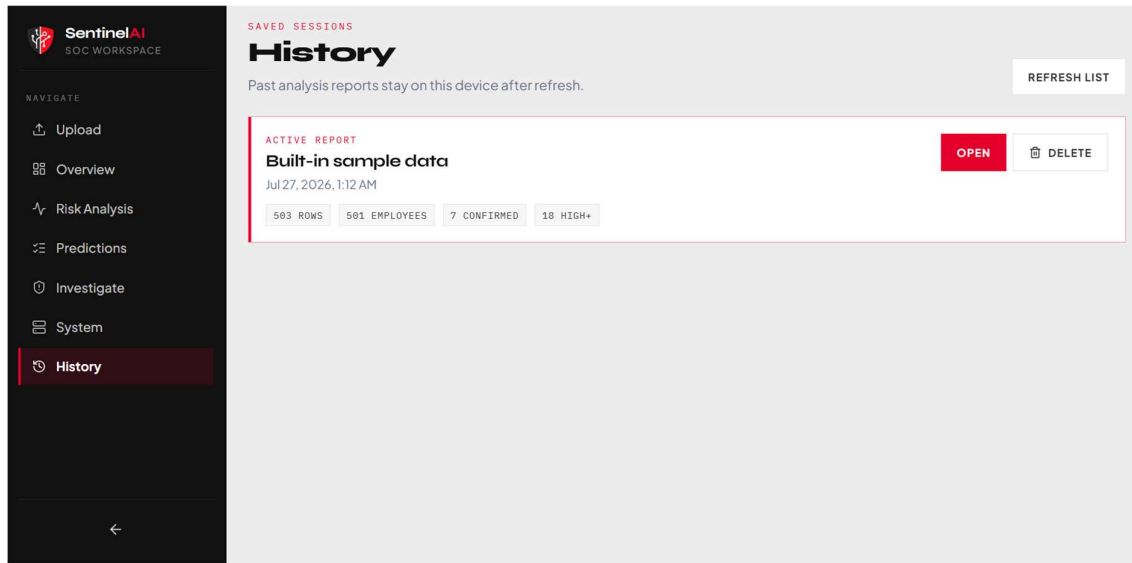
4.9 History Module

The History Module stores previous analyses, allowing analysts to revisit earlier investigations and compare historical security trends. This feature supports audit requirements and long-term behavioural monitoring.

Features

- Previous Analysis Records
- Search and Filter
- Historical Risk Comparison
- Case Reopening

Figure 4.9 History Module



CHAPTER 5

SYSTEM ARCHITECTURE

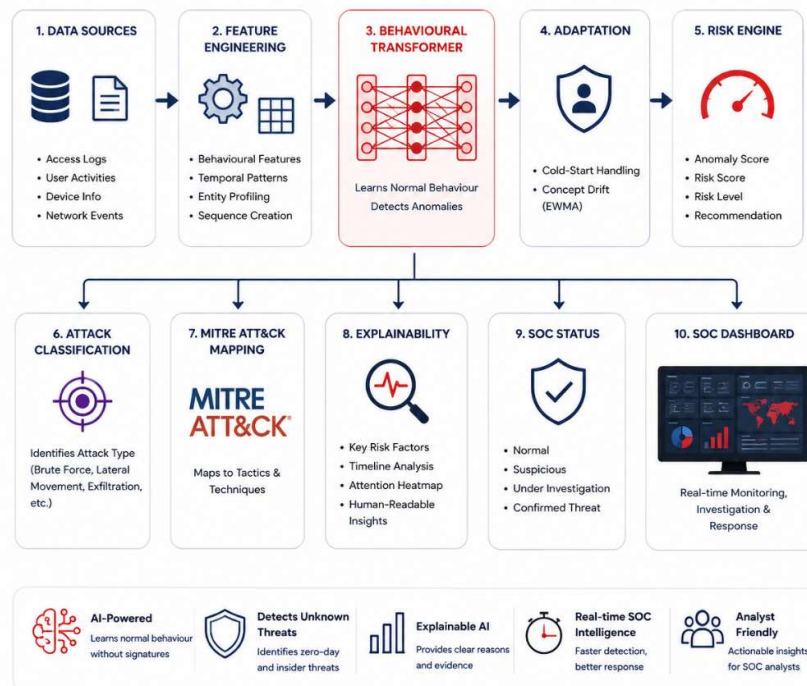
The SentinelAI architecture consists of multiple interconnected modules that collectively perform behavioural anomaly detection, risk assessment, and incident investigation.

The system begins by collecting enterprise security logs, which may originate from authentication systems, VPN gateways, endpoint devices, application servers, and cloud services. These logs undergo feature engineering to transform raw events into structured behavioural feature vectors and sequential activity records.

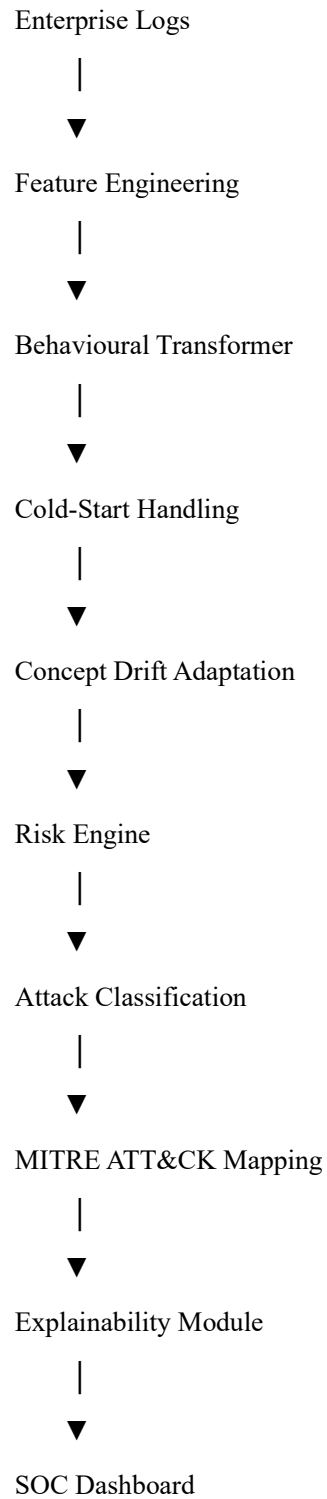
The Behavioural Transformer processes these sequences to determine how much a user's activity deviates from learned normal behaviour. The resulting anomaly score is then refined through cold-start adjustment and concept drift adaptation before being passed to the Risk Engine.

The Risk Engine combines anomaly scores with deterministic cybersecurity rules to calculate a final risk score and classify the severity of the incident. The Attack Classification module determines the most probable attack type, while the MITRE Mapping module associates the incident with appropriate ATT&CK tactics and techniques. Finally, the Explainability module produces analyst-friendly summaries, contributing factors, timelines, and recommendations, all of which are displayed through the React-based SOC Dashboard.

Figure 5.1 Overall System Architecture



5.1 Architectural Workflow



CHAPTER 6

TECHNOLOGY STACK

The SentinelAI platform is developed using modern technologies across machine learning, backend development, frontend visualization, and cybersecurity analytics.

Layer	Technology Used
Programming Language	Python 3.11
Machine Learning	PyTorch
Data Processing	Pandas, NumPy
Synthetic Data Generation	Faker
Backend Framework	FastAPI
API Validation	Pydantic
Web Server	Uvicorn
Frontend	React 19
Language	TypeScript
Build Tool	Vite
Data Visualization	Recharts
UI Animation	Framer Motion
API Communication	Axios
Testing	PyTest
Documentation	Swagger UI

6.1 Development Environment

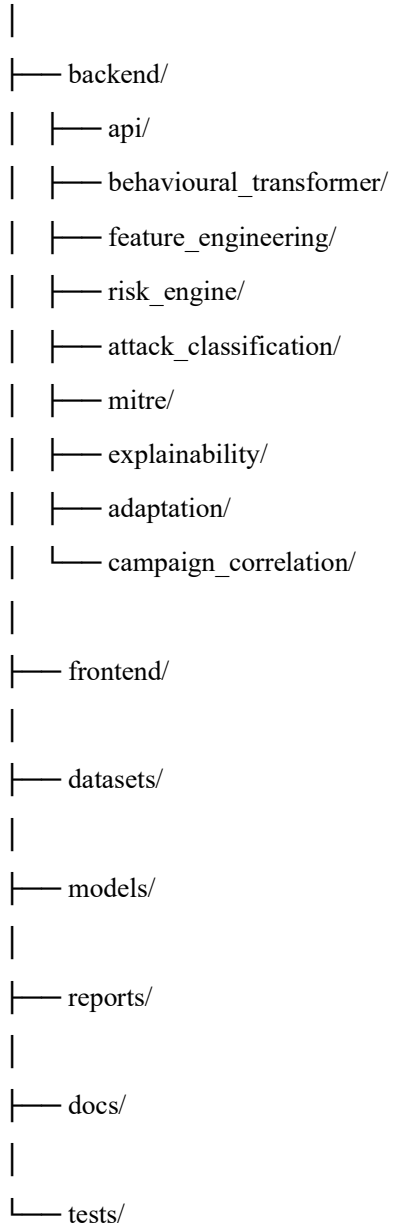
The backend of SentinelAI is implemented using FastAPI, providing RESTful APIs for prediction, batch analysis, streaming data, campaign correlation, and health monitoring. PyTorch is used to implement the Behavioural Transformer model responsible for anomaly detection. Feature engineering and synthetic data generation are handled using Pandas, NumPy, and Faker.

The frontend is developed using React 19 with TypeScript to create an interactive Security Operations Center dashboard. Charts and analytics are displayed using Recharts, while Axios facilitates communication between the frontend and backend services. Swagger UI provides interactive API documentation, allowing developers and analysts to test endpoints directly through the browser.

6.2 Repository Structure

The project repository is organised into multiple modules, each responsible for a specific aspect of the cybersecurity pipeline.

SentinelAI/



This modular architecture ensures maintainability, scalability, and ease of future enhancements.

CHAPTER 7

METHODOLOGY

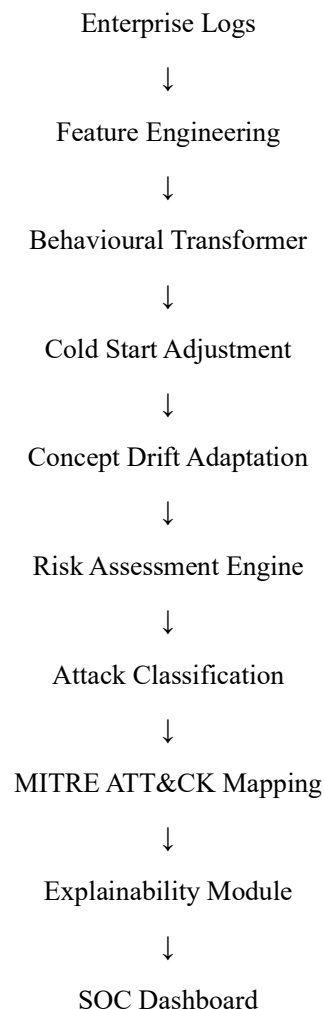
7.1 Overview

SentinelAI follows a multi-stage behavioural cybersecurity pipeline designed to identify sophisticated cyber threats by learning normal enterprise behaviour rather than relying on predefined attack signatures. The complete methodology consists of data collection, feature engineering, behavioural modelling, anomaly detection, risk assessment, attack classification, MITRE ATT&CK mapping, explainability generation, and Security Operations Center (SOC) visualization.

Unlike traditional machine learning systems that directly classify attack labels, SentinelAI intentionally separates anomaly detection from attack classification. The Behavioural Transformer learns behavioural patterns in an unsupervised manner without exposure to attack labels, thereby preventing label leakage. Deterministic cybersecurity rules are then applied to interpret the anomaly and assign an appropriate risk level.

The complete workflow is illustrated below.

Figure 7.1 Complete Methodology



CHAPTER 8

FEATURE ENGINEERING

Feature engineering transforms raw enterprise events into structured behavioural representations suitable for machine learning. Instead of analysing isolated events, SentinelAI aggregates a user's daily activity into a behavioural profile known as a **Feature Vector**.

Each feature vector captures important behavioural characteristics, including login frequency, authentication failures, geographical movement, resource access diversity, VPN usage, session duration, command execution patterns, download activity, and endpoint device changes. These engineered features provide meaningful context for distinguishing normal behaviour from suspicious activity.

To preserve the integrity of anomaly detection, attack labels are excluded from the machine learning feature set. Ground-truth labels are retained only for evaluation purposes and never influence the Behavioural Transformer during inference.

Key Features

- Authentication failure rate
- Number of login attempts
- Device switching frequency
- Geographic location changes
- Resource access entropy
- Session duration
- VPN usage ratio
- File download volume
- Burst activity detection
- Command sequence patterns

The generated feature vectors are combined with sequential event data to provide a comprehensive behavioural representation for each employee.

CHAPTER 9

BEHAVIOURAL TRANSFORMER

9.1 Overview

The Behavioural Transformer is the primary anomaly detection component of SentinelAI. It is implemented using the PyTorch deep learning framework and is trained exclusively on normal behavioural sequences. Unlike supervised classifiers, the model does not learn attack labels. Instead, it learns the underlying structure of legitimate enterprise behaviour.

During inference, the model reconstructs the observed behavioural sequence. A high reconstruction error indicates that the behaviour differs significantly from previously learned patterns and is therefore considered anomalous.

This unsupervised learning strategy enables SentinelAI to identify previously unseen threats, insider attacks, and zero-day behaviours without requiring labelled attack datasets.

9.2 Transformer Architecture

The Behavioural Transformer consists of three major components:

1. Input Embedding Layer

- Converts behavioural features into numerical embeddings suitable for sequence processing.

2. Transformer Encoder

- Utilizes multi-head self-attention to capture relationships between behavioural events across time.
- Learns long-range dependencies within user activity sequences.

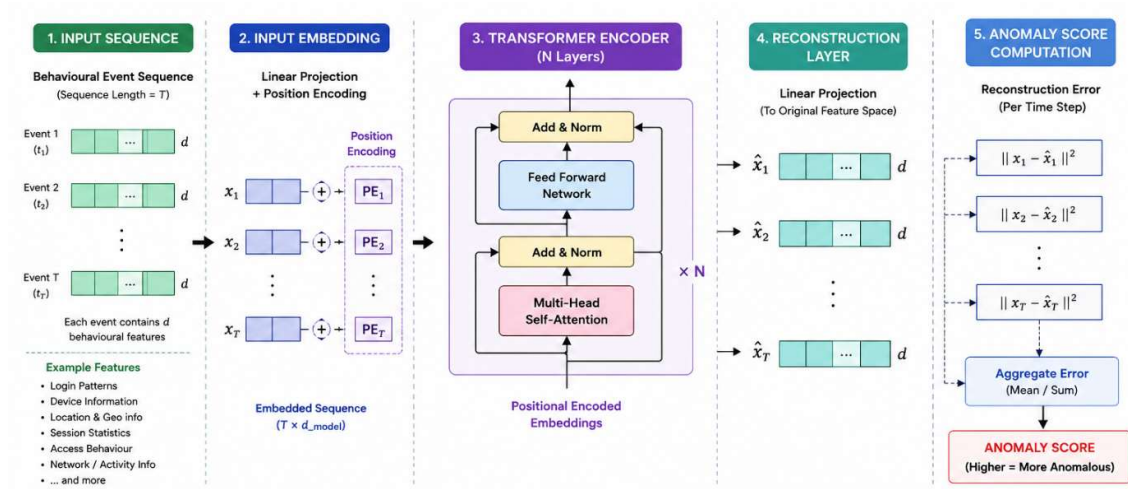
3. Reconstruction Layer

- Reconstructs the original behavioural sequence.
- Computes reconstruction error used as the anomaly score.

Typical configuration:

Parameter	Value
Sequence Length	64
Embedding Dimension	64
Attention Heads	4
Encoder Layers	3
Framework	PyTorch

Figure 9.1 Behavioural Transformer Architecture



9.3 Why a Transformer?

Traditional machine learning models analyse independent observations and often fail to capture temporal relationships. In contrast, the Transformer architecture models the sequential nature of enterprise behaviour, allowing SentinelAI to understand how user actions evolve over time.

For example, a login followed by privileged resource access and large data downloads forms a meaningful behavioural sequence that may indicate an attack. By modelling these dependencies, the Transformer achieves superior behavioural understanding compared to static classifiers.

CHAPTER 10

COLD-START HANDLING

One of the major challenges in behavioural cybersecurity is the **cold-start problem**, where newly observed users or devices possess insufficient historical data for reliable anomaly detection. Directly applying anomaly scores to such entities may result in inaccurate risk assessments.

SentinelAI addresses this issue by introducing a Cold-Start Adjustment module. The module identifies entities with limited behavioural history and adjusts their anomaly scores toward a neutral prior. This prevents unnecessary alerts while allowing the model to gradually learn the behavioural profile of new users.

Each prediction includes an `is_cold_start` flag, enabling SOC analysts to distinguish between genuinely suspicious behaviour and uncertainty caused by limited historical information.

Benefits

- Reduces false positives for new employees.
- Improves analyst confidence.
- Enables gradual behavioural learning.
- Supports scalable enterprise deployment.

CHAPTER 11

CONCEPT DRIFT ADAPTATION

User behaviour naturally evolves over time due to role changes, new responsibilities, remote work, or organisational restructuring. Static anomaly detection models often misclassify these legitimate behavioural changes as malicious activity.

SentinelAI incorporates a Concept Drift Adaptation module based on Exponentially Weighted Moving Average (EWMA). The module continuously monitors historical anomaly scores for each entity.

If behavioural changes occur gradually, the anomaly score is dampened, indicating likely behavioural evolution. Conversely, sudden spikes in anomaly scores are preserved, signalling potential cyber intrusions requiring immediate investigation.

This adaptive mechanism ensures that SentinelAI remains accurate even as enterprise behaviour changes over long periods.

CHAPTER 12

RISK ASSESSMENT ENGINE

The Risk Assessment Engine converts anomaly scores into actionable cybersecurity intelligence. Instead of relying solely on the machine learning output, the engine combines behavioural anomalies with deterministic security indicators to generate a final enterprise risk score.

The following behavioural signals contribute to the final risk calculation:

- Multiple authentication failures
- Impossible geographic travel
- Excessive file downloads
- VPN misuse
- Unusual working hours
- High resource access entropy
- Device changes
- Burst activity

Each indicator contributes to the overall risk score.

Risk Levels

Score	Risk Level
-------	------------

0–24	LOW
------	-----

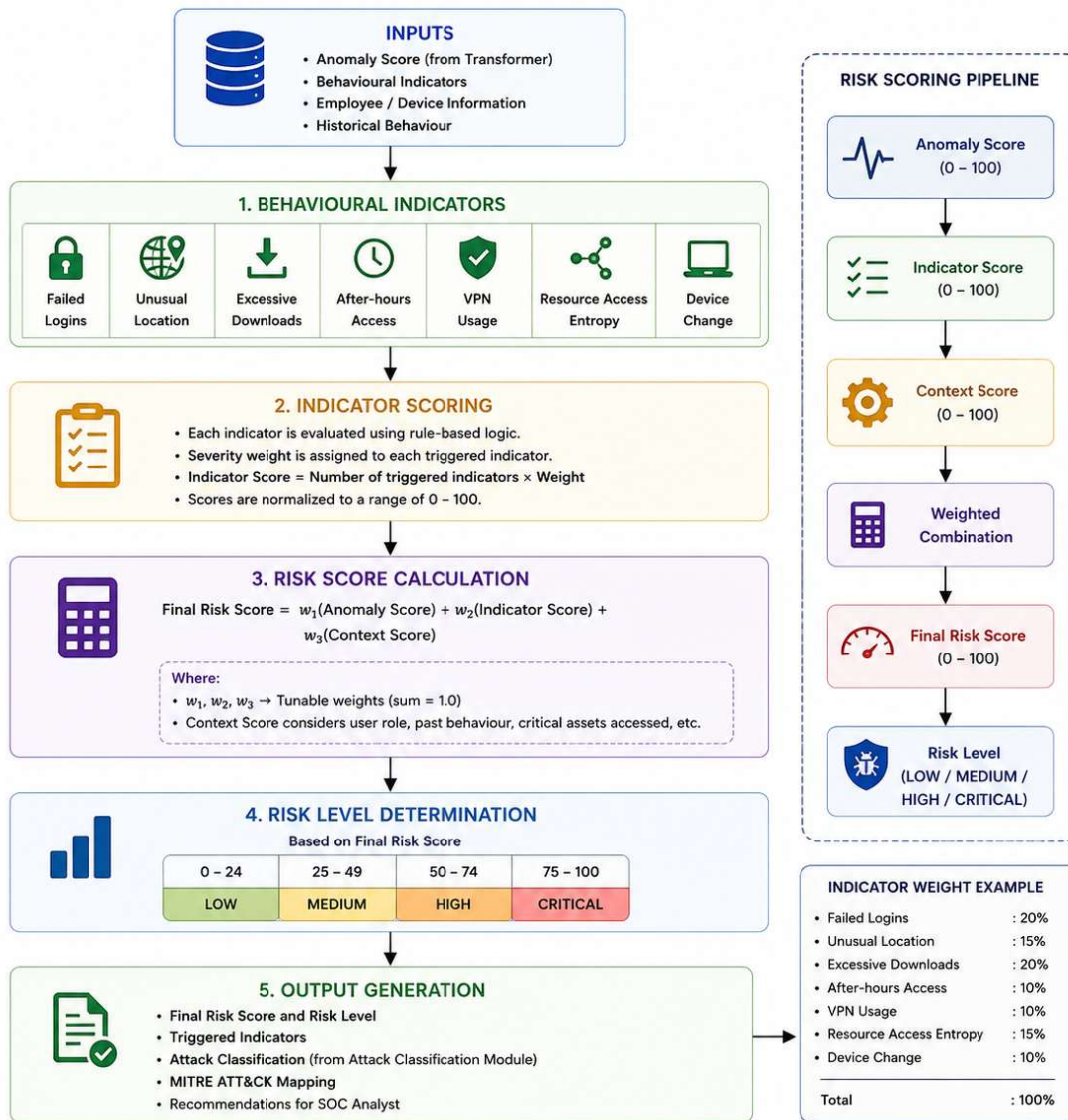
25–49	MEDIUM
-------	--------

50–74	HIGH
-------	------

75–100	CRITICAL
--------	----------

The engine also generates security recommendations to guide SOC analysts during incident response.

Figure 12.1 Risk Assessment Workflow



CHAPTER 13

ATTACK CLASSIFICATION

13.1 Overview

After the Behavioural Transformer identifies an anomalous behavioural pattern and the Risk Assessment Engine calculates the corresponding risk score, SentinelAI determines the most probable cyberattack category. Unlike many machine learning-based security solutions, attack classification is performed using deterministic cybersecurity rules rather than neural network predictions.

This separation ensures that behavioural anomaly detection remains independent of attack labels, preventing label leakage while making the final decision transparent and explainable.

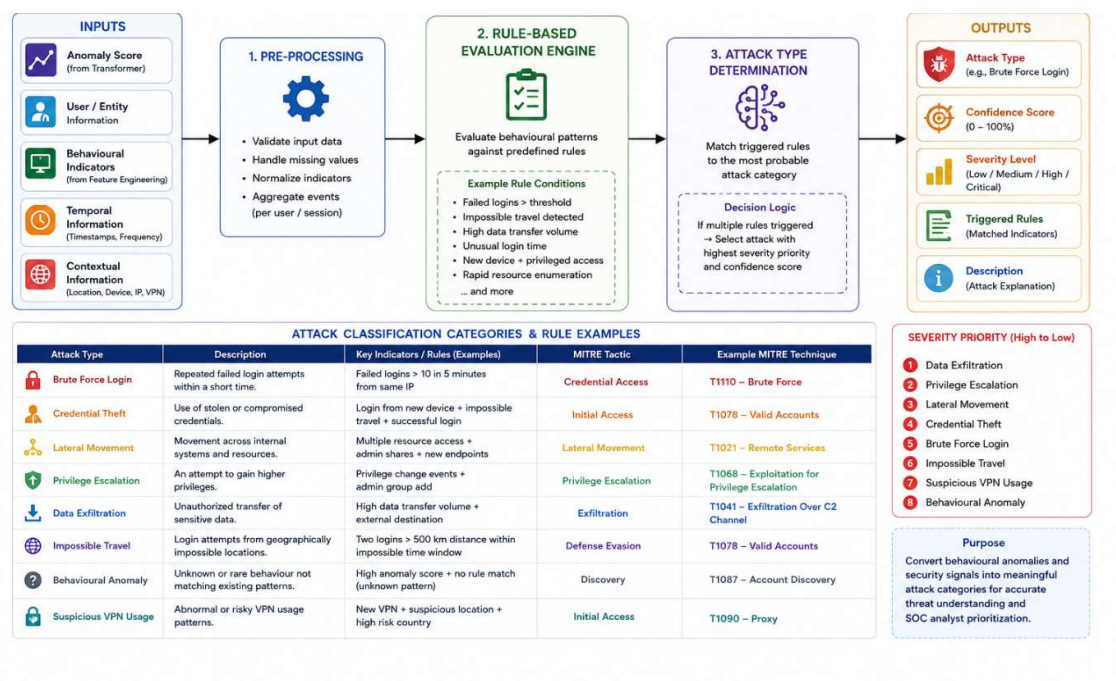
The attack classification module analyses several behavioural indicators, including authentication failures, resource access diversity, device changes, geographic movement, abnormal download activity, VPN usage, and session behaviour. Based on predefined cybersecurity logic, each anomaly is assigned to the most appropriate attack category.

The platform currently supports the following attack categories:

- Brute Force Login
- Credential Theft
- Credential Stuffing
- Privilege Escalation
- Lateral Movement
- Impossible Travel
- Device Spoofing
- Data Exfiltration
- Low-and-Slow Exfiltration
- Suspicious VPN Usage
- Insider Activity
- Behavioural Anomaly (Unknown Pattern)
- Normal Activity

This rule-based approach enables analysts to understand exactly why a particular alert has been assigned to a specific attack category.

Figure 13.1 Attack Classification Workflow



CHAPTER 14

MITRE ATT&CK MAPPING

The MITRE ATT&CK framework is a globally recognised knowledge base that categorises adversarial tactics and techniques observed in real-world cyberattacks. Integrating MITRE ATT&CK mapping into SentinelAI enables SOC analysts to understand attacks using standardised cybersecurity terminology.

After identifying the attack type, SentinelAI automatically maps it to the corresponding MITRE tactic and technique.

Sample MITRE Mapping

Attack Type	MITRE Technique	Tactic
Brute Force Login	T1110	Credential Access
Credential Theft	T1078	Initial Access
Lateral Movement	T1021	Lateral Movement
Privilege Escalation	T1068	Privilege Escalation
Impossible Travel	T1078	Valid Accounts
Data Exfiltration	T1041	Exfiltration
Low-and-Slow Exfiltration	T1030	Exfiltration
Device Spoofing	T1090	Defense Evasion

Providing this mapping helps analysts quickly understand attacker behaviour, prioritise investigations, and align responses with industry-standard cybersecurity practices.

CHAPTER 15

EXPLAINABILITY MODULE

15.1 Overview

Artificial Intelligence systems often function as black boxes, making it difficult for analysts to understand why a prediction was made. SentinelAI addresses this challenge through a dedicated Explainability Module that converts technical model outputs into analyst-friendly explanations.

Instead of displaying only an anomaly score, the system provides a comprehensive explanation that includes the behavioural factors responsible for the alert, the corresponding attack classification, risk level, MITRE ATT&CK mapping, and recommended response actions.

The explainability engine combines insights from the Behavioural Transformer with deterministic rule-based observations to generate transparent and actionable reports.

15.2 Explanation Components

Each prediction includes the following:

- Executive summary
- Behavioural anomaly score
- Risk score and severity level
- Triggered behavioural indicators
- Identified attack type
- MITRE ATT&CK mapping
- Recommended analyst actions
- Timeline of suspicious events
- Attention-weighted behavioural importance

This information enables analysts to understand the reasoning behind every alert and reduces the time required for incident investigation.

CHAPTER 16

KILL-CHAIN CORRELATION

One of the distinguishing features of SentinelAI is its ability to correlate multiple suspicious events occurring over several days into a single attack campaign. Traditional detection systems often generate isolated alerts, requiring analysts to manually determine whether they are related.

SentinelAI introduces a Kill-Chain Correlation module that links sequential attack stages associated with the same employee or device. This capability enables analysts to view an entire attack lifecycle rather than individual alerts.

For example, an attacker may first perform repeated failed logins, then successfully authenticate using stolen credentials, later move laterally across the network, and finally download confidential information. Individually, these events may appear harmless; collectively, they reveal a coordinated attack campaign.

By correlating related events, SentinelAI reduces investigation effort, improves contextual awareness, and helps analysts respond more effectively to advanced persistent threats.

CHAPTER 17

IMPLEMENTATION

The backend of SentinelAI is implemented using FastAPI, exposing RESTful APIs for prediction, batch analysis, streaming data processing, campaign correlation, and system health monitoring. The Behavioural Transformer model is trained offline and loaded during application startup, ensuring efficient inference without retraining during prediction.

The frontend is developed using React 19 and TypeScript, providing an intuitive Security Operations Center dashboard. Analysts can upload datasets, monitor enterprise-wide risk, investigate suspicious cases, and visualise attack patterns through interactive charts and tables.

The system workflow is as follows:

1. Upload enterprise logs or use sample data.
2. Perform feature engineering.
3. Generate anomaly scores using the Behavioural Transformer.
4. Apply cold-start and concept drift adjustments.
5. Calculate risk score and classify attack type.
6. Map attacks to MITRE ATT&CK.
7. Generate explainability outputs.
8. Display results through the SOC dashboard.

CHAPTER 18

RESULTS AND EVALUATION

The performance of SentinelAI was evaluated using a synthetic enterprise dataset containing both normal and malicious behavioural patterns. Evaluation focused on the system's ability to prioritise suspicious activities for Security Operations Center analysts while maintaining low false positive rates.

Evaluation Metrics

- Precision
- Recall
- F1 Score
- Top-k Alert Accuracy
- False Positive Rate
- Risk Classification Accuracy

The Behavioural Transformer effectively learned normal enterprise behaviour and successfully identified anomalous activity without relying on labelled attack data. Cold-start handling reduced false alerts for newly observed entities, while concept drift adaptation enabled the system to accommodate gradual behavioural changes.

The kill-chain correlation module further enhanced investigation efficiency by linking related incidents into unified attack campaigns.

Figure 18.1 Confusion Matrix

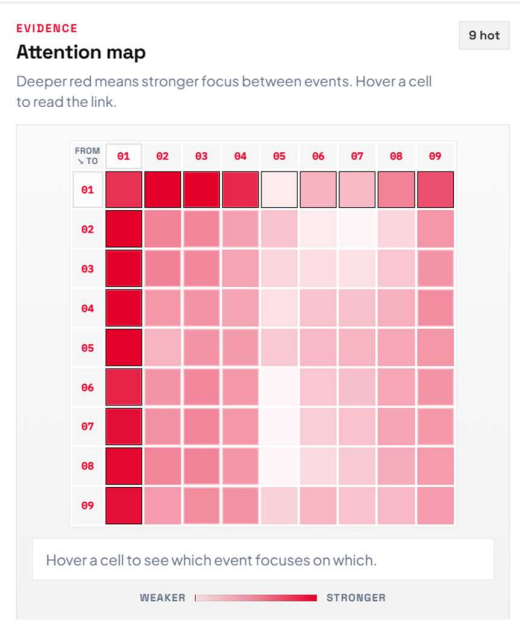
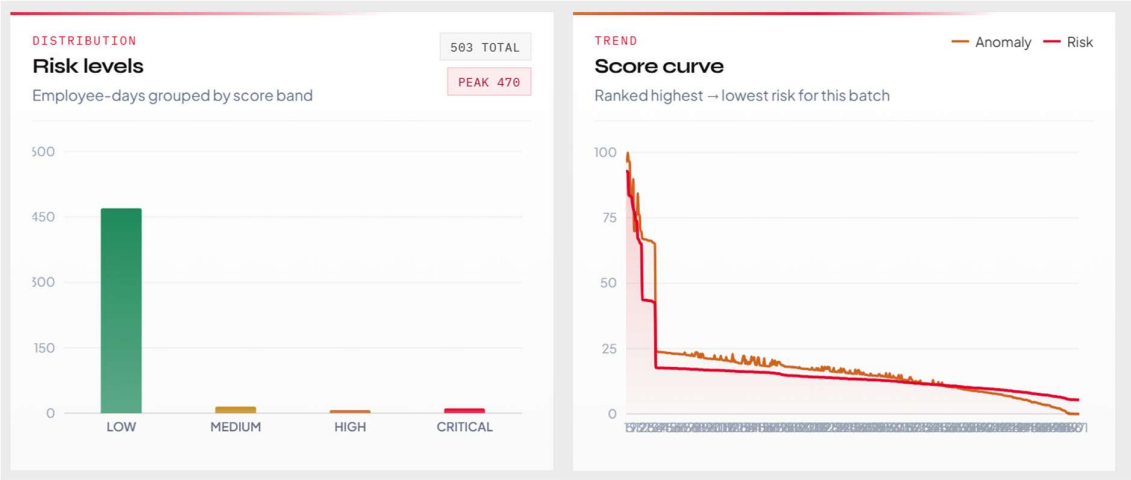


Figure 18.2 Risk Distribution



CHAPTER 19

ADVANTAGES & FUTURE SCOPE

19.1 Advantages

The proposed SentinelAI platform offers several significant advantages:

- Detects unknown cyber threats without predefined signatures.
- Uses unsupervised learning to model normal behaviour.
- Prevents label leakage by separating anomaly detection from attack classification.
- Supports explainable AI for transparent decision-making.
- Handles behavioural evolution through concept drift adaptation.
- Addresses the cold-start problem for new users and devices.
- Correlates multi-stage attacks into complete campaigns.
- Provides an interactive SOC dashboard for efficient incident investigation.
- Supports scalable deployment using FastAPI and React.

19.2 Future Scope

Several enhancements can further improve SentinelAI:

- Integration with enterprise SIEM platforms such as Splunk and Microsoft Sentinel.
- Real-time Kafka or RabbitMQ streaming for large-scale deployments.
- Multi-cloud monitoring across AWS, Azure, and Google Cloud Platform.
- Integration with Security Orchestration, Automation, and Response (SOAR) platforms.
- Automatic incident response through playbook execution.
- Federated learning for privacy-preserving behavioural modelling.
- Large Language Model (LLM) integration for automated incident summaries and analyst assistance.
- Mobile SOC dashboard for remote monitoring.

CHAPTER 20

CONCLUSION

SentinelAI demonstrates how Artificial Intelligence can transform enterprise cybersecurity by shifting the focus from signature-based detection to behavioural intelligence. The platform combines unsupervised deep learning, deterministic risk analysis, explainable AI, and interactive visualisation to provide a comprehensive Security Operations Center solution.

By learning normal behavioural patterns, detecting anomalies, classifying attack types, mapping incidents to the MITRE ATT&CK framework, and correlating multi-stage campaigns, SentinelAI enables faster, more accurate, and more transparent cyber threat detection. The incorporation of cold-start handling, concept drift adaptation, and explainability makes the system practical for dynamic enterprise environments.

The modular architecture, based on FastAPI and React, ensures scalability and maintainability while supporting future integration with real-world enterprise infrastructure. SentinelAI therefore represents a modern, intelligent, and extensible approach to behavioural cybersecurity, aligning with the objectives of the Honeywell Hackathon.

CHAPTER 21

REFERENCES

1. MITRE ATT&CK Framework. <https://attack.mitre.org/>
2. Vaswani, A., et al. *Attention Is All You Need*. NeurIPS, 2017.
3. PyTorch Documentation. <https://pytorch.org/>
4. FastAPI Documentation. <https://fastapi.tiangolo.com/>
5. React Documentation. <https://react.dev/>
6. OWASP Foundation. <https://owasp.org/>
7. Microsoft Security Documentation. <https://learn.microsoft.com/security/>
8. NIST Cybersecurity Framework. <https://www.nist.gov/cyberframework>

APPENDIX

GitHub Repository

<https://github.com/eshitaguptaa/SentinelAI-Behavioral-Threat-Detection>

Deployed Site

<https://sentinel-ai-behavioral-threat-detection-keedprxpu.vercel.app/>

Demo Video

<https://drive.google.com/drive/folders/1lx4J1kTKugMUaBTTGAO9ncda5uuYa4Xy?usp=sharing>